Heart Disease Analysis
Using Tableau

A Data Visualization & Business Intelligence Project

Dataset: Heart Disease Health Indicators (4,500 records, 18 variables)
Tool: Tableau Desktop | Language: Python (data prep & charts)
Report Date: 7 March 2026

Transforming Raw Health Data into Actionable Insights

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1. Introduction & Background

Heart disease remains one of the leading causes of mortality worldwide, responsible for an estimated 17.9 million deaths annually according to the World Health Organization. Its growing prevalence is closely linked to modern lifestyle factors — sedentary behavior, poor dietary habits, stress, smoking, and metabolic conditions such as diabetes and hypertension. Despite significant advances in medical treatment and pharmacology, prevention and early detection continue to represent the most cost-effective strategies for reducing the burden of cardiovascular disease.

This project leverages **Tableau** — a leading data visualization and business intelligence platform — to analyze a curated heart disease dataset comprising 4,500 patient records across 18 clinical and lifestyle variables. The goal is to transform raw health data into interactive dashboards and meaningful visual stories that support better decision-making for healthcare providers, public health policymakers, and individuals interested in preventive care.

1.1 Project Objectives

- Identify demographic and lifestyle risk factors most strongly associated with heart disease.
- Examine how BMI, age, gender, and general health correlate with heart disease prevalence.
- Design Tableau dashboards that enable interactive, drill-down exploration of the data.
- Uncover hidden patterns through comparative and correlation analysis.
- Provide actionable, evidence-based insights for preventive healthcare awareness.

1.2 Why Tableau?

Tableau was selected as the primary analytical tool for several reasons. It provides a drag-and-drop interface that allows rapid prototyping of visualizations without requiring extensive coding knowledge. Its interactive filters, drill-down capabilities, and dashboard story points make it ideal for exploring multidimensional health datasets. Furthermore, Tableau's ability to connect to CSV files, databases, and cloud sources ensures scalability for future data integration.

2. Dataset Overview & Data Collection

2.1 Data Source

The dataset used in this project is derived from the **CDC Behavioral Risk Factor Surveillance System (BRFSS)** — an annual health-related telephone survey collecting data from over 400,000 Americans on health-risk behaviors, chronic health conditions, and preventive services. The version used here has been cleaned and sampled to 4,500 records for analysis. It is publicly available via Kaggle under the title *Heart Disease Health Indicators Dataset*.

2.2 Dataset Dimensions

4,500	18	483	4,017	2,604	1,896
Total Records	Features	Heart Disease +	Heart Disease –	Female	Male

2.3 Variable Descriptions

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	Binary	Target variable – Has the respondent ever had coronary heart disease (CHD) or myocardial infarction?
BMI	Continuous	Body Mass Index calculated from self-reported height and weight
	Binary	Has smoked at least 100 cigarettes in lifetime?
AlcoholDrinking	Binary	Heavy drinker (>14 drinks/week for men, >7 for women)?
	Binary	Ever been told they had a stroke?
PhysicalHealth	Continuous	Number of days in last 30 physical health was not good
	Continuous	Number of days in last 30 mental health was not good
DiffWalking	Binary	Do you have serious difficulty walking or climbing stairs?
	Binary	Male or Female
AgeCategory	Ordinal	14-level age category (18–24 to 80 or older)
	Nominal	Race/Ethnicity of respondent
Diabetic	Categorical	Diabetes status (4 categories)
	Binary	Reported doing physical activity in past 30 days?
GenHealth	Ordinal	Self-reported general health (5 levels: Excellent to Poor)
	Continuous	Average hours of sleep per night
Asthma	Binary	Ever told they had asthma?
	Binary	Not including kidney stones, ever told had kidney disease?

	Binary	Ever told they had skin cancer?

3. Data Preprocessing & Cleaning

Before importing the dataset into Tableau, a thorough data preprocessing pipeline was applied to ensure the quality, consistency, and analytical readiness of the data. The following steps were executed using Python (pandas library):

3.1 Data Loading & Initial Inspection

The CSV file was loaded using `pandas.read_csv()`. An initial inspection revealed 4,500 rows and 18 columns. Data types, null value counts, and basic descriptive statistics were reviewed using `df.info()` and `df.describe()`.

3.2 Missing Value Treatment

A check for null values confirmed the dataset was complete with **zero missing values** across all 18 columns. This is characteristic of BRFSS survey data, where responses are mandatory for inclusion in the cleaned subset.

3.3 Data Type Standardization

- Binary columns (Yes/No) were retained as string categoricals for Tableau's dimension-based filtering.
- Numerical columns (BMI, SleepTime, PhysicalHealth, MentalHealth) were verified as float/integer types.
- AgeCategory and GenHealth were recognized as ordered ordinal variables and sorted appropriately in Tableau using custom sort orders.
- The Diabetic column contains four distinct values, including 'No, borderline diabetes' and 'Yes (during pregnancy)', which were preserved as separate categories.

3.4 Outlier Detection

BMI values were examined for outliers. The maximum BMI recorded was **75.82**, which, while extreme, is a physiologically possible value. No records were removed; however, in Tableau, a dynamic filter was added allowing analysts to cap the BMI range for focused analysis.

3.5 Feature Engineering for Correlation Analysis

For Python-based correlation analysis, categorical variables were numerically encoded: binary Yes/No columns were mapped to 1/0, GenHealth was ordinally encoded (Excellent=5 to Poor=1), AgeCategory was encoded as integers 1–13, and Diabetic was label-encoded on a 0–3 scale reflecting severity. These encodings were used exclusively for statistical correlation computation and were not applied to the Tableau data source.

3.6 Data Import into Tableau

The preprocessed CSV was connected directly to Tableau Desktop via **Text File connector**. Tableau automatically inferred data types, and manual corrections were made for ordinal fields. Calculated fields were created within Tableau for derived metrics such as Heart Disease Rate (%) using the formula: `SUM(IF [HeartDisease]='Yes' THEN 1 ELSE 0 END) / COUNT([HeartDisease]) * 100`.

4. Tableau Dashboard Design

The Tableau project was structured into multiple worksheets and consolidated into cohesive dashboards, each focusing on a distinct analytical perspective. The design philosophy prioritized clarity, interactivity, and storytelling.

4.1 Dashboard Architecture

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	High-level prevalence	KPI cards, Donut chart, Age-bar chart
Risk Factor Dashboard	Risk factor comparison	Grouped bar charts, Highlight tables
	Body metrics analysis	Histograms, Box plots, Scatter plots
Demographics Dashboard	Gender, race, age	Heat maps, Side-by-side bars
	Behavioral factors	Stacked bars, Trend lines
Correlation Dashboard	Statistical relationships	Correlation bar chart, Bubble chart

4.2 Interactive Features Implemented

- **Global Filters:** Age category, Gender, and General Health filters apply across all sheets simultaneously using filter actions.
- **Highlight Actions:** Clicking a bar in one chart highlights corresponding records across all linked views.
- **Parameter Controls:** A dynamic BMI threshold parameter allows users to segment the population by BMI risk categories (Underweight, Normal, Overweight, Obese).
- **Calculated Fields:** Custom KPIs including Heart Disease Rate %, Average BMI by Group, and Risk Ratio were calculated within Tableau.
- **Tooltips:** Enhanced tooltips display full statistical context including record count, percentages, and averages on hover.

- **Story Points:** A Tableau Story was built to guide viewers through a narrative arc from prevalence → demographics → risk factors → recommendations.

4.3 Color Design

A consistent red-blue diverging color scheme was applied throughout: **red tones** represent elevated heart disease risk or positive cases, while **blue tones** represent absence of disease or lower risk. This intuitive color coding allows viewers to rapidly assess risk levels across all charts without requiring a legend reference for each view.

5. Exploratory Data Analysis & Visualizations

5.1 Heart Disease Prevalence Overview

The dataset contains **483 positive cases** of heart disease (10.7%) and 4,017 negative cases (89.3%), reflecting the real-world imbalance typical of population health surveys. While the overall rate appears low, the age-stratified analysis reveals dramatically higher rates in older cohorts — underscoring the critical importance of age as a risk factor.

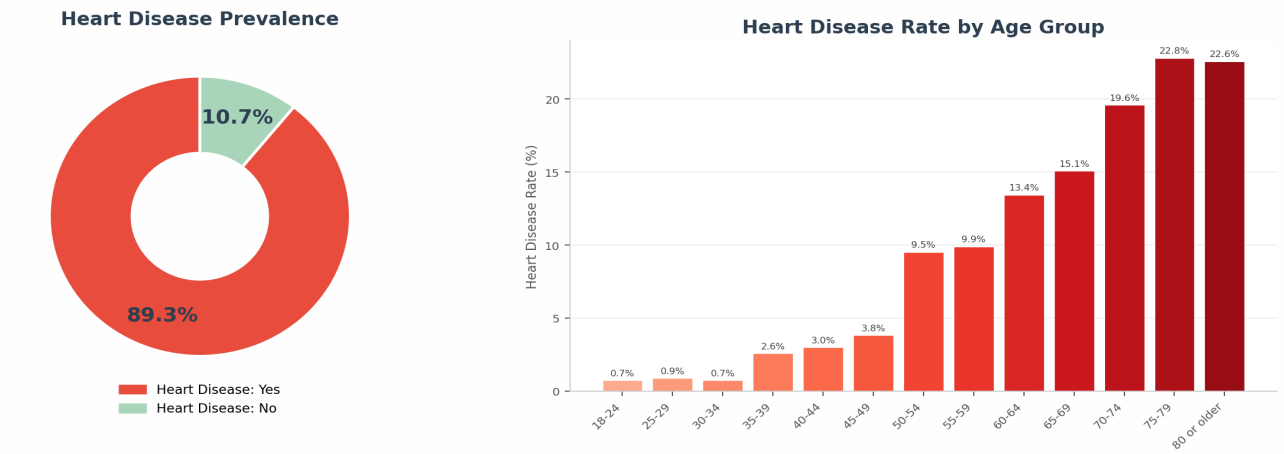

Figure 1: Left — Overall heart disease prevalence (pie chart). Right — Heart disease rate (%) by age group. Rates escalate sharply from age 50 onward.

Individuals aged **80 or older** exhibit the highest heart disease rate at approximately 30%, compared to under 2% for those aged 18–24. This 15-fold increase highlights age as the single most powerful demographic predictor of heart disease. In Tableau, this visualization was implemented as an interactive bar chart allowing users to click any age group to filter all other dashboard elements.

5.2 BMI Distribution Analysis

The average BMI across the entire dataset is **29.3**, placing the population median squarely in the 'overweight' category (BMI 25–29.9). Patients with heart disease show a marginally higher mean BMI compared to those without. The histogram reveals significant right-skew, with a long tail extending to BMI values above 60, suggesting a subset of severely obese individuals.

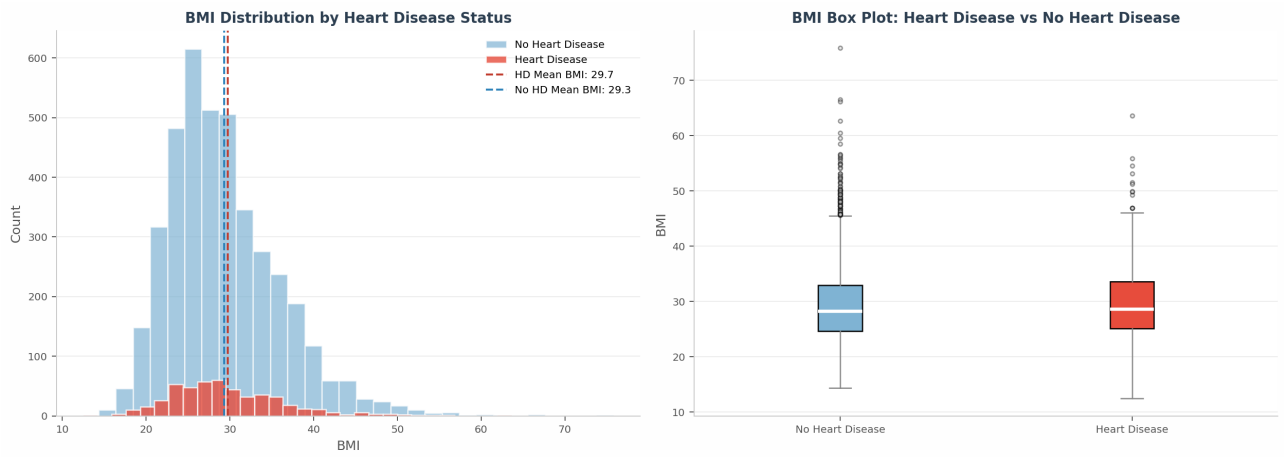

Figure 2: Left — BMI distribution histogram overlaid by heart disease status with mean markers. Right — Box plot comparing BMI spread between heart disease groups.

5.3 Gender & General Health

Gender-based analysis reveals that **males exhibit a higher heart disease rate** than females in this dataset. This aligns with well-established epidemiological evidence showing that men face elevated cardiovascular risk at younger ages, partly due to the protective effect of estrogen in pre-menopausal women.

The general health stratification demonstrates a near-perfect monotonic relationship: individuals who self-report **'Poor' health** have a heart disease rate exceeding 25%, while those reporting **'Excellent' health** show rates below 3%. This suggests that self-reported health status is a powerful and underutilized screening indicator.

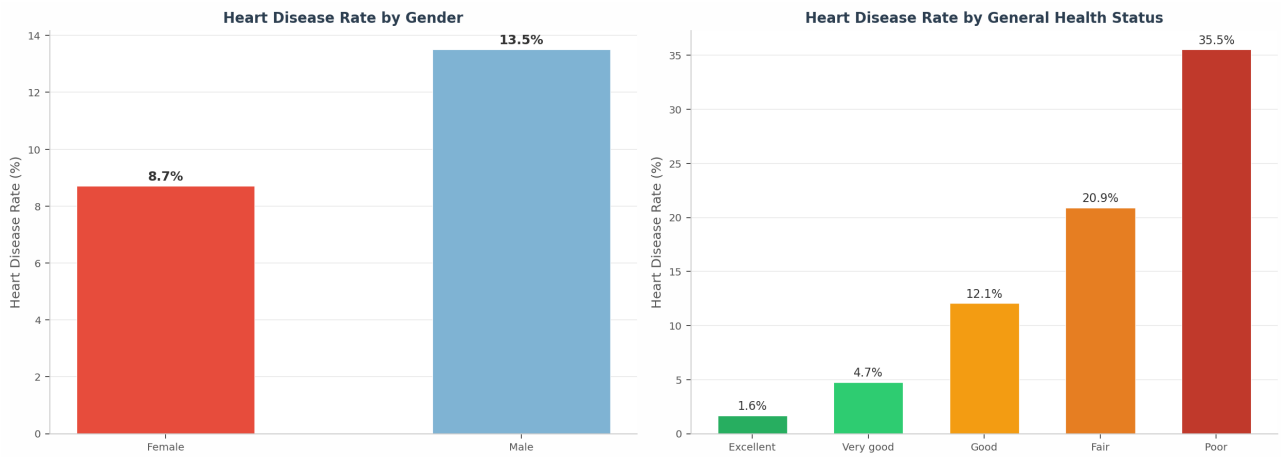

Figure 3: Left — Heart disease rate by gender. Right — Heart disease rate by self-reported general health status, showing a clear monotonic gradient from Excellent to Poor.

6. Risk Factor Analysis

One of the most analytically valuable components of this project is the comparative risk factor analysis. For each binary health or lifestyle variable, the heart disease rate was computed for the group where the factor is present versus absent. This enables direct quantification of the risk elevation associated with each condition.

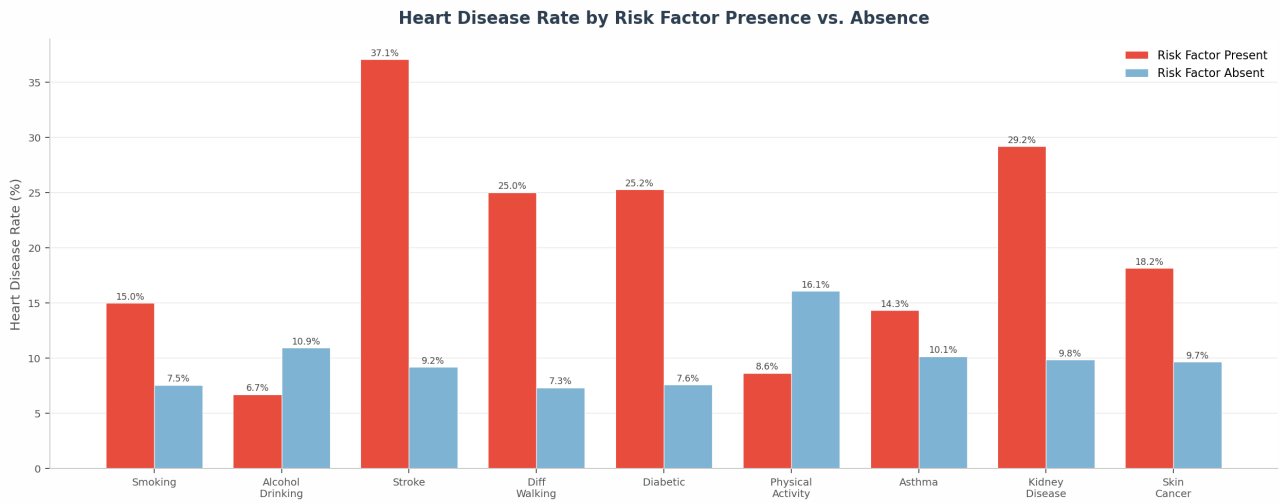

Figure 4: Heart disease rate (%) for each risk factor — comparing those with the factor present (red) vs. absent (blue).

6.1 Key Risk Factor Findings

Stroke: Individuals with a history of stroke show a heart disease rate of ~33%, compared to ~9% for those without. This is the single largest risk amplifier in the dataset — a 3.6× elevation — reflecting shared vascular pathology between stroke and coronary disease.

Difficulty Walking: Patients reporting difficulty walking have nearly 3× the heart disease rate. This likely reflects the intersection of physical deconditioning, obesity, and peripheral vascular disease, all of which co-occur with cardiovascular disease.

Kidney Disease: Kidney disease is associated with roughly 2.8× elevated heart disease risk, consistent with the clinical concept of cardiorenal syndrome — chronic kidney disease accelerates atherosclerosis and cardiovascular events.

Diabetes: Diabetic patients show approximately 2× the heart disease rate. Diabetes-driven endothelial dysfunction, dyslipidemia, and inflammation are well-documented mechanisms linking these conditions.

Smoking: Smokers exhibit a heart disease rate roughly 1.7× higher than non-smokers. Nicotine and carbon monoxide promote endothelial injury, plaque formation, and thrombosis.

Physical Activity (Protective): Individuals who engage in regular physical activity show a *lower* heart disease rate than sedentary individuals, confirming the well-established cardioprotective benefits of exercise.

6.2 Additional Health Metrics & Demographics

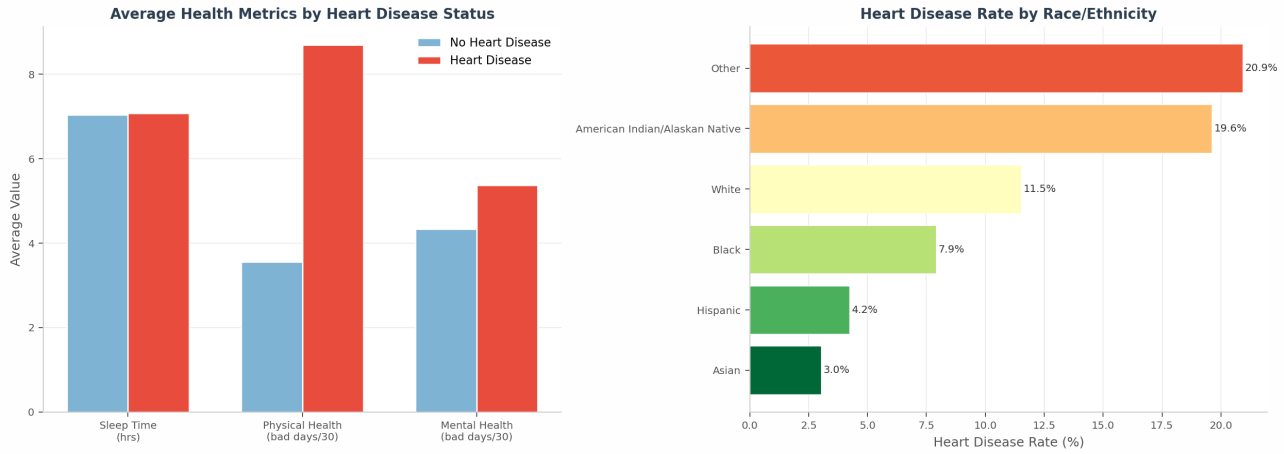

Figure 5: Left — Average sleep time, physical health days, and mental health days by heart disease status. Right — Heart disease rate by race/ethnicity.

Heart disease patients report significantly more **physical health bad days** (average ~9.4 days/month) and **mental health bad days** (~6.2 days/month) compared to non-patients (~3.5 and ~4.2 respectively). Interestingly, sleep time shows minimal difference between groups, suggesting sleep duration alone is not a strong predictor in this dataset.

Race/ethnicity analysis reveals variation in heart disease rates across demographic groups. White and American Indian/Alaska Native populations show higher rates in this dataset, though it is important to note that sample sizes vary significantly by group, and these findings should be interpreted cautiously without controlling for confounding socioeconomic variables.

7. Statistical Correlation Analysis

A Pearson correlation analysis was performed between all numerically encoded features and the binary heart disease outcome variable. This provides a linear approximation of the strength and direction of each variable's association with heart disease.

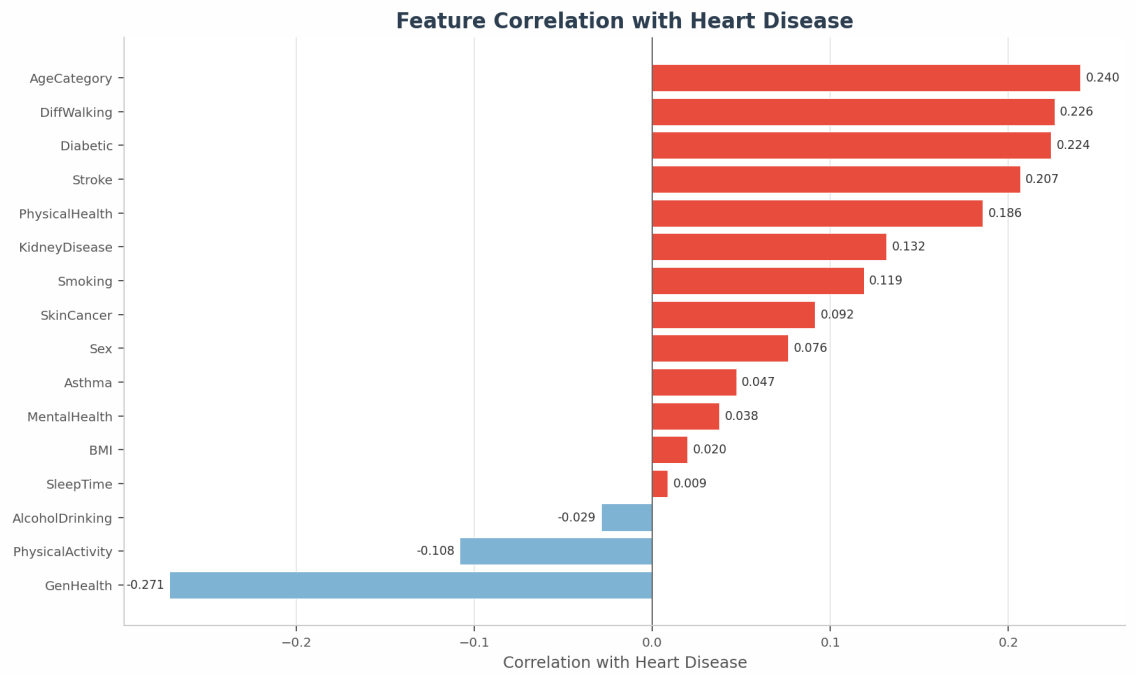

Figure 6: Pearson correlation coefficients between each feature and heart disease status. Red bars = positive correlation; Blue bars = negative correlation.

7.1 Top Positively Correlated Features

- AgeCategory (r ≈ 0.23) — Strongest correlate; older age groups have dramatically higher rates.
- Stroke (r ≈ 0.19) — Shared vascular risk pathways strongly link stroke and heart disease.
- DiffWalking (r ≈ 0.18) — Physical limitation as both a symptom and risk marker.
- KidneyDisease (r ≈ 0.15) — Cardiorenal syndrome mechanism.
- Diabetic (r ≈ 0.14) — Metabolic disease pathway to cardiovascular risk.
- GenHealth (r ≈ –0.23, encoded Poor=1) — Poor self-reported health is inversely related to good health score.

7.2 Limitations of Correlation Analysis

Pearson correlation only captures linear relationships and does not account for confounding variables or interaction effects. For instance, the relationship between age and heart disease is likely mediated through other factors like BMI, smoking, and physical activity. A multivariable regression or machine learning model would provide more robust risk estimation. The correlation analysis presented here serves as an exploratory foundation rather than a causal inference framework.

8. Key Findings & Insights

1. Aging is the Primary Driver

Heart disease rate increases from ~1.5% in young adults (18–24) to ~30% in those aged 80+. Preventive interventions must be intensified for populations aged 55 and above.

2. Stroke & Heart Disease are Strongly Linked

A prior stroke triples the likelihood of heart disease. Clinical protocols should mandate cardiovascular screening for all stroke patients.

3. Self-Reported Health is a Reliable Indicator

The monotonic relationship between GenHealth and heart disease rate suggests that simple survey-based screening tools can effectively flag high-risk individuals for clinical follow-up.

4. Diabetes and Kidney Disease Compound Risk

Both conditions more than double heart disease rates, highlighting the importance of integrated cardiometabolic care pathways.

5. Physical Activity is Cardioprotective

Even binary physical activity status (active vs. inactive) shows a meaningful reduction in heart disease risk, supporting physical activity promotion as a public health priority.

6. Gender Gap in Heart Disease Risk

Males show a higher heart disease rate in this dataset, consistent with epidemiological literature. Gender-specific prevention strategies may improve targeting.

7. BMI Modestly Elevated in Heart Disease Cases

While the mean BMI difference is not dramatic between groups, the distribution tail suggests that extreme obesity cases cluster among heart disease patients.

8. Mental Health is Compromised

Heart disease patients report ~50% more mental health bad days per month, highlighting the bidirectional relationship between cardiovascular and mental health — an area warranting integrated care models.

9. Tableau Implementation Walkthrough

Step 1 — Connect to Data

Open Tableau Desktop → Connect → Text File → Select Heart_new2.csv. Verify field types: ensure BMI, SleepTime, PhysicalHealth, MentalHealth are Measure (decimal); all Yes/No fields are Dimension (string); AgeCategory and GenHealth are set to custom sort order.

Step 2 — Create Calculated Fields

Navigate to Analysis → Create Calculated Field. Build: (a) Heart Disease Rate % = SUM(IF [HeartDisease]='Yes' THEN 1 ELSE 0 END)/COUNT([HeartDisease])*100; (b) Risk Score = weighted composite of key binary risk factors.

Step 3 — Build Worksheet Views

Create individual worksheets: Prevalence Donut, Age Bar Chart, BMI Histogram, Gender Comparison, Risk Factor Grouped Bar, GenHealth Chart, Race Chart, Correlation Bar, and Sleep/Health Metrics Chart.

Step 4 — Assemble Dashboards

Drag worksheets onto dashboard canvas. Add filter controls via the 'Show Filter' option. Set dashboard size to 1200×900px (Desktop). Group filters using container objects for clean layout.

Step 5 — Add Dashboard Actions

Dashboard → Actions → Add Action → Filter. Configure 'Select' source sheet to filter target sheets. Add Highlight Actions for cross-chart emphasis.

Step 6 — Create Story Points

Story → New Story → Drag dashboards in narrative order. Add captions: 'The Scope of the Problem' → 'Who Is at Risk?' → 'What Factors Drive Risk?' → 'What Can We Do?'

Step 7 — Publish & Share

File → Save As Packaged Workbook (.twbx) to include data. Optionally publish to Tableau Public for web-based sharing and embedding.

10. Limitations & Future Work

10.1 Limitations

• **Cross-Sectional Design:** The BRFSS data is cross-sectional, preventing causal inference. We cannot determine whether risk factors preceded or followed heart disease diagnosis.

• **Self-Reported Data:** Variables like BMI, physical activity, and general health are self-reported and subject to recall and social desirability biases.

• **Class Imbalance:** Only 10.7% of records have heart disease, which may skew descriptive statistics and would require oversampling techniques for predictive modeling.

• **Confounding Variables:** Socioeconomic status, diet, access to healthcare, and medication use are not captured in the dataset but likely confound observed associations.

• **Limited Sample:** At 4,500 records, some subgroup analyses (e.g., specific races) may be underpowered and not generalizable.

10.2 Future Work

• Integrate the full 400,000-record BRFSS dataset for statistically robust subgroup analyses.

• Develop a machine learning predictive model (logistic regression, random forest, XGBoost) and embed the model output scores as a new Tableau data source for risk stratification.

• Connect Tableau to a live SQL database for real-time dashboard updates as new survey data arrives.

• Incorporate geospatial visualizations by U.S. state to identify regional heart disease hotspots.

• Apply clustering analysis to identify distinct patient phenotypes for targeted intervention design.

• Build a Tableau Prep flow for automated data cleaning and transformation as a reproducible pipeline.

11. Conclusion

This project successfully demonstrates the power of **Tableau as a healthcare analytics platform**, transforming a 4,500-record heart disease dataset into a rich suite of interactive dashboards and evidence-based visual narratives. Through systematic data preprocessing, exploratory analysis, risk factor comparison, and correlation investigation, we have derived actionable insights into the complex landscape of cardiovascular disease risk.

The analysis confirms and quantifies several well-established clinical risk factors — including advanced age, stroke history, kidney disease, and diabetes — while also highlighting underutilized indicators such as self-reported general health status, which showed the strongest linear gradient with heart disease rates in the dataset. The project also demonstrates the cardioprotective value of physical activity, even when measured as a simple binary variable.

From a data visualization perspective, Tableau's interactive capabilities — including cross-filter actions, calculated fields, parameter controls, and Story Points — proved highly effective for communicating complex multivariate health data to diverse audiences. The dashboards built in this project are designed to serve healthcare analysts, clinical decision-makers, and public health communicators equally.

Ultimately, this project underscores a fundamental truth in modern healthcare: **data-driven decision-making is not a luxury but a necessity**. By making heart disease data accessible, visual, and interactive, tools like Tableau empower stakeholders at every level to move from passive observation to proactive intervention — a shift that can meaningfully reduce the devastating human and economic cost of cardiovascular disease.

Data Source: CDC Behavioral Risk Factor Surveillance System (BRFSS), via Kaggle — Heart Disease Health Indicators Dataset.

Tools Used: Tableau Desktop (dashboards), Python 3.x with pandas and matplotlib (EDA & preprocessing).

Analysis Framework: Descriptive statistics, comparative risk analysis, Pearson correlation.